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ENERGY

The Work Already Done

Enrichment is measured in a unit that was never properly published, and the unit explains why the hard part comes first.

3,280 words · 17-minute read

Some units have a founding document. The metre has a treaty, the ampere has a definition adopted by acclamation at a General Conference, and the joule has Joule. The unit in which the world buys and sells uranium enrichment has a note that Paul Dirac wrote in 1941 and never properly published.

Dirac had spent the early part of the war on isotope separation, and the idea that lasted from it was an answer to a question separation engineers could not otherwise answer: what, exactly, does an enrichment plant produce? Not the mass of its product, because turning natural uranium into reactor fuel and turning reactor fuel into bomb metal are wildly different amounts of work for the same output mass. Not energy, because energy depends on the technology and the whole point was to compare technologies. Dirac proposed a quantity he called separative work, and Klaus Fuchs and Rudolf Peierls adopted it in a classified report the following year [1]. Karl Cohen's 1951 volume in the National Nuclear Energy Series is the first systematic open treatment [2]. The unit is now written SWU, pronounced "swoo", and about sixty million of them are sold every year.

The Unit Dirac Never Published

Start with the bookkeeping. An enrichment plant takes in feed of mass F at U-235 weight fraction x_F and splits it into product of mass P at the higher assay x_P and tails, or waste, of mass T at the lower assay x_T . Uranium is conserved and so is U-235, which gives two equations and hence, for a chosen product, everything:

$$F = P \frac{x_P - x_T}{x_F - x_T}, \quad T = P \frac{x_P - x_F}{x_F - x_T}.$$

Natural uranium is 0.711 per cent U-235 by weight; a light-water reactor wants three to five per cent. For one kilogram of five per cent product with tails discarded at 0.25 per cent, those equations say the plant must swallow 10.30 kilograms of natural uranium and throw away 9.30 kilograms of nearly-depleted material. Enrichment is overwhelmingly an exercise in disposal.

Dirac's contribution was to assign to any uranium stream a value depending only on its assay,

$$V(x) = (2x - 1) \ln\left(\frac{x}{1 - x}\right),$$

and to define the work done by a separation as the value it creates:

$$\Delta U = PV(x_P) + TV(x_T) - FV(x_F).$$

One separative work unit is one kilogram of that quantity, which is the first oddity: because the value function is dimensionless, a SWU has the dimensions of mass, and plant capacity is quoted in kilograms per year even though nothing about it is a mass flow.

The left panel of the figure below draws V . It vanishes at $x = 1/2$, is symmetric about that point, and rises logarithmically towards both ends. It is not a price and not an energy; it measures how far a mixture sits from being half-and-half, in the units that make separative work additive. A kilogram of natural uranium scores 4.87 and a kilogram of ninety per cent bomb metal scores 1.76, which sounds absurd until you remember that the function is applied to masses, and that the masses here are lopsided beyond intuition.

Run the arithmetic for the five per cent kilogram and it comes to 7.92 SWU. The World Nuclear Association's worked examples give 7.9 SWU for that case, 4.8 SWU for 3.5 per cent product at the same tails assay, and 4.3 SWU if the tails rise to 0.30 per cent [3]. The formula reproduces all three to the digit quoted — 7.923, 4.811 and 4.339.

Additive, and Route-Independent

Suppose you want a kilogram of ninety per cent uranium from natural feed, with a tails assay of 0.25 per cent. You could run one enormous cascade in a single pass. Or make 4.95 per cent reactor fuel first and feed that into a second, smaller cascade. Or stop at 19.75 per cent, or at sixty. Do these cost different amounts of separative work?

They do not. Provided the intermediate stage returns its tails at the natural assay — where they are simply recycled as feed to the first stage, which is the physically honest way to make the comparison — every route requires exactly the same separative work, the same natural uranium, and leaves the same tails. Checking six intermediate assays from two per cent to sixty against the single-pass cascade returns 208.03186 SWU per kilogram every time, agreeing to 1.7 parts in 10^{13} , which is machine precision rather than approximation. Separative work is a potential. Only the endpoints matter.

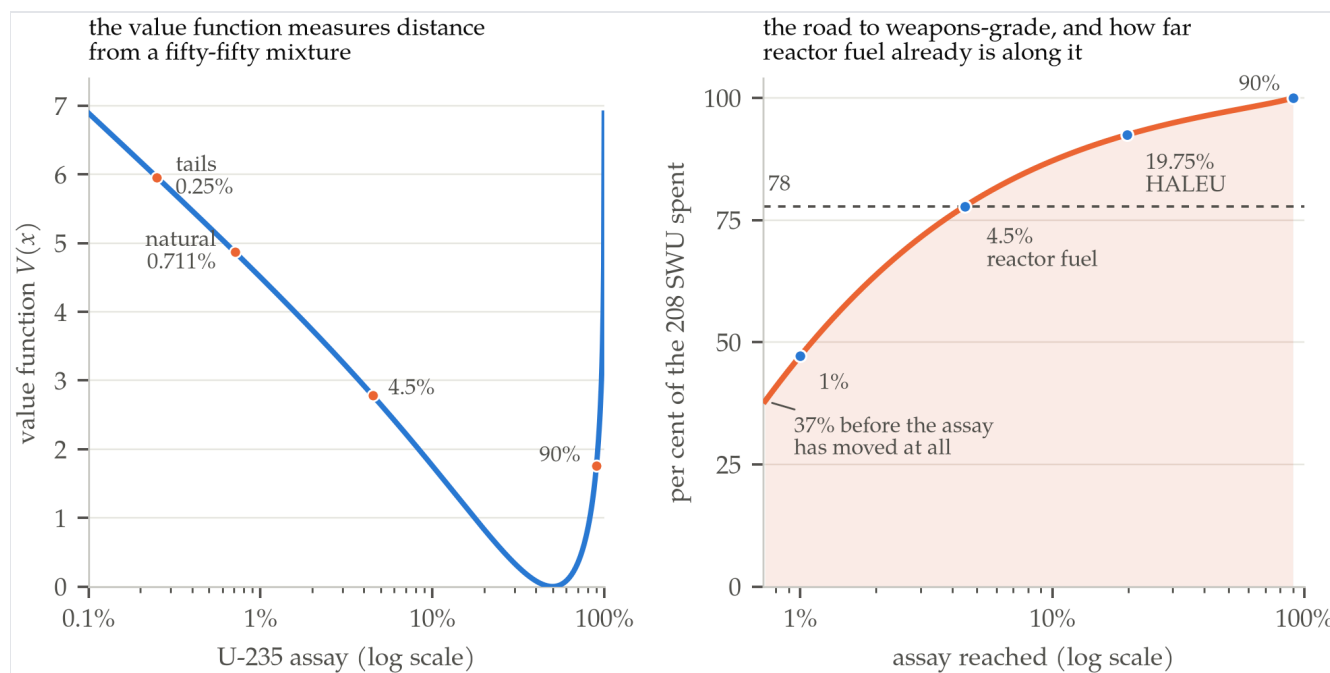


FIGURE Left: Dirac's value function on a logarithmic assay axis. It vanishes at a fifty-fifty mixture and diverges at both ends, so the natural-uranium assay of 0.711 per cent scores far higher than ninety per cent bomb metal does. Right: for one kilogram of ninety per cent product drawn from natural feed at 0.25 per cent tails — 208.03 SWU in total — the share of that work already spent by the time the material reaches a given assay. Because the total is route-independent, the split is well defined. Reactor fuel at 4.5 per cent sits at 78 per cent of the way.

That is a pleasant mathematical fact with an unpleasant consequence, which the right panel draws. Because the work is route-independent, it makes sense to ask how much of the 208 SWU has been spent by the time the material reaches a given assay — and almost all of it is spent early.

OF THE JOURNEY FROM NATURAL URANIUM TO WEAPONS-GRADE, WHICH HALF IS THE EXPENSIVE HALF?

The first, and not by a small margin. By the time the material reaches one per cent U-235 — a change so slight that no reactor would accept it as fuel — 47 per cent of the separative work has been done. By 4.5 per cent, the assay of ordinary light-water reactor fuel, it is 78 per cent. By 19.75 per cent, the ceiling of the high-assay low-enriched uranium being built for advanced reactor designs, 92 per cent. By sixty per cent, 98. The last stretch from sixty to ninety, the one that sounds like the difficult part, is four SWU out of 208.

The reason is the lopsided mass. One kilogram of ninety per cent material takes 194.7 kilograms of natural uranium, but only 21.1 kilograms of 4.95 per cent, or 4.7 kilograms of 19.75 per cent. Separative work scales with the tonnage pushed through the machines, and the tonnage collapses as the assay rises. The final cascade in a two-stage route is a small machine handling a trickle; the first is a factory.

There is even a floor. Push the intermediate assay down towards natural uranium and the share does not fall to zero but converges to 37 per cent, because stripping tails to 0.25 per cent out of a very large feed stream is work every route must pay before the product assay has moved at all.

Why a Diffusion Plant Needs Twelve Hundred Stages

None of that says anything about how you separate. It is a statement about the task, deliberately technology-blind, which is exactly what Dirac wanted. The machines enter through a single number: the separation factor α by which one pass through one stage multiplies the abundance ratio $x/(1-x)$ of a stream.

For gaseous diffusion, in which uranium hexafluoride is pushed against a porous barrier and the lighter molecules get through very slightly faster, the ideal separation factor is the square root of the mass ratio of the two molecules:

$$\alpha_{\text{ideal}} = \sqrt{\frac{352.04}{349.03}} = 1.00430.$$

Four parts in a thousand. In an ideal cascade, where the streams that meet at each junction have matching assays and no separative work is wasted on remixing, the abundance ratio climbs by $\sqrt{\alpha}$ per stage, so the stage count between two assays is a logarithm divided by a very small number:

$$N = \frac{\ln(R_P/R_T)}{\ln \sqrt{\alpha}}, \quad R = \frac{x}{1-x}.$$

Route	Diffusion, 1.0043	Centrifuge, 1.3	Centrifuge, 1.5
Natural to 3 per cent	682	11	7
Full cascade to 3 per cent	1,172	19	12
Natural to 4.4 per cent	868	14	9
Full cascade to 90 per cent	3,817	62	40

The second row is the one to check against reality, because the published figure for a diffusion plant making ordinary three per cent reactor fuel is about 1,200 stages [4]. The formula gives 1,172 once the stripping section is counted — the stages below the feed point that chase the assay down to the tails specification — and 682 if only the enriching section is. That the published number matches the whole cascade rather than the enriching half is a structural fact the arithmetic hands over for free: two fifths of the stages in a diffusion plant exist to make the waste more wasteful.

A centrifuge changes α by two orders of magnitude and the stage count collapses with it, to nineteen. Why is a neat piece of dimensional reasoning. In a long counter-current centrifuge the separation depends on the pressure gradient rotation establishes across the radius, which goes as the square of the peripheral speed, and the separative power on the square of that — so it scales as the fourth power of rotor speed and only linearly with rotor length [6]. Dirac derived that limit too, which is why a centrifuge programme is really a materials programme: everything is bought with tip speed, and tip speed is set by the tensile strength of whatever you can spin without it flying apart. Aluminium gave way to maraging steel and then to carbon fibre for that reason and no other. The law is, it turns out, too

generous: Tronin and colleagues showed in 2021 that the derivation ignores the vacuum core that forms around the axis at high speeds, where no separation happens at all, and that the corrected limit grows only as the square of the speed [5].

The engineering that made this practical belongs largely to Gernot Zippe, an Austrian engineer captured at the end of the war and put to work on the Soviet programme at Sukhumi, where he solved the problem that had defeated everyone else: supporting a rapidly spinning rotor for years without the bearing destroying itself, which he did by pairing a needle pivot with a magnetic bearing. Released in 1956 and intercepted by American intelligence, he spent two years at the University of Virginia reproducing from memory what he had built in Georgia, and a version of that machine is the ancestor of nearly every centrifuge now turning [7]. The modern industry descends from a bearing.

What Three Thousand Megawatts Bought

The most vivid way to feel the difference between the two technologies is the electricity bill. Gaseous diffusion consumes roughly 2,400 kilowatt-hours per SWU; a modern centrifuge plant, about fifty [3].

The Paducah Gaseous Diffusion Plant in Kentucky, which enriched uranium from 1952 to 2013, drew about 3,000 megawatts from the Tennessee Valley Authority — enough, as the Department of Energy puts it, to power a city the size of Nashville [8]. Run 3,040 megawatts, the reported peak demand of its four process buildings, for a year and you get 26.6 terawatt-hours; divide by 2,400 kilowatt-hours per SWU and you get 11.1 million SWU a year. Paducah's design capacity was 11.3 million. Three numbers from three separate places — a peak electrical demand, a process energy intensity, and a plant rating — agree to within two per cent, which is about as good a confirmation as a figure gets without opening the paper it came from.

Put that same 26.6 terawatt-hours through centrifuges and it would deliver 533 million SWU a year, roughly eight and a half times the entire world's present enrichment capacity of about 61 million. One obsolete plant's electricity, redirected through modern machines, would enrich fuel for eight planets.

Run the comparison the other way. A kilogram of 4.5 per cent enriched uranium taken to 45 gigawatt-days per tonne of burnup at 33 per cent thermal efficiency yields 356 megawatt-hours of electricity, and enriching it takes 6.87 SWU, or 0.34 megawatt-hours — an energy return on the enrichment step of about 1,040 to one. Under gaseous diffusion that step cost 16.5 megawatt-hours and the return was twenty-two to one, still positive but a different industry.

RUNNING IT BACKWARDS

There is one large historical case of the process in reverse. Under the 1993 Megatons to Megawatts agreement, Russia downblended five hundred tonnes of weapons-grade uranium from dismantled warheads — the equivalent of some twenty thousand of them — into low-enriched fuel for American reactors. The programme ran twenty years, concluded in December 2013, and over its life supplied close to half of all American nuclear fuel, which is to say roughly ten per cent of all American electricity [9].

In separative-work terms, blending down is vandalism on a heroic scale. Five hundred tonnes of ninety per cent material embodies 104 million SWU relative to natural uranium, or one year and nine months of total world enrichment output at present capacity, and dilution destroys all of it. Whether it was a good trade is not a close question. But the unit invented to price the industry also prices the disarmament, and prices it very high.

Choosing How Much to Throw Away

The tails assay x_T has been sitting in every formula above as though it were given. It is not. It is a decision, and the most consequential discretionary number in the fuel cycle.

Lower tails mean squeezing more U-235 out of each kilogram of feed: less uranium, more separative work. Higher tails mean the opposite. Since the plant buys both, the question is a minimisation. Writing r for the price of a kilogram of feed divided by the price of one SWU, the cost per kilogram of product comes to $rF + \Delta U$ and setting its derivative in x_T to zero gives a condition on r alone:

$$r = V(x_F) - V(x_T) - (x_F - x_T) V'(x_T).$$

Two things about that expression deserve attention. The first is that it is correct: it is the condition quoted in the trade literature, and differentiating the cost function symbolically returns it exactly rather than approximately. The second is what is missing from it. The product assay x_P does not appear, so the optimal tails assay depends only on the price ratio and the feed assay, and not at all on what you are making. A plant enriching to 4.5 per cent for a pressurised water reactor and one enriching to 19.75 per cent for a small modular reactor should, at the same prices, discard material of identical quality.

The condition is monotone and steep at the cheap-uranium end. At $r = 0.5$ the optimal tails assay is 0.302 per cent, at $r = 1$ it is 0.227, at $r = 2$ it is 0.158, and at $r = 3$ it falls to 0.124 — so a plant facing cheap uranium and dear centrifuges throws away material two and a half times richer than one facing the reverse.

At the end of August 2026 the long-term price of uranium concentrate was about 96 dollars a pound, conversion about 53 dollars per kilogram of uranium, and enrichment about 181 dollars per SWU [10]. A pound of triuranium octoxide contains 0.3846 kilograms of uranium, so delivered feed costs about 303 dollars per kilogram and r is about 1.67, implying an optimal tails assay of 0.175 per cent. Real plants run higher, typically 0.20 to 0.30 per cent, because contracts cap the tails assay a supplier may use without a surcharge and because the naive optimum ignores the cost of carrying inventory through a cascade with a long equilibrium time [11].

The interesting part is what has not moved. Spot separative work traded near 56 dollars in 2021 and above 200 in mid-2026, a consequence of the sanctions that followed February 2022 and of the American prohibition on imported Russian low-enriched uranium that took effect in August 2024. But uranium

tripled too, from the low thirties to the high eighties a pound. The ratio r went from about 1.88 to about 1.67 and the optimal tails assay from 0.164 to 0.175 per cent. A crisis that transformed the industry barely touched its most important operating parameter, because it moved both prices at once.

The leverage in that parameter is nonetheless large. Against world reactor requirements of about 68,900 tonnes of uranium in 2025 [12], shifting the whole industry from 0.20 to 0.30 per cent tails would raise natural uranium demand by roughly 14,800 tonnes a year, more than a fifth, while freeing about 12 million SWU — a fifth of world enrichment capacity. Uranium and centrifuges are substitutes, and the exchange rate between them is a curve any fuel buyer can differentiate.

The Cost of Fuel, and the Cost of One Bomb

Assembling the stack at 2026 long-term prices, for one kilogram of 4.5 per cent fuel at 0.20 per cent tails, gives a bill of about 4,240 dollars.

Component	Quantity	Cost, dollars	Share
Natural uranium	8.41 kgU	2,100	50 per cent
Conversion	8.41 kgU	446	11 per cent
Enrichment	7.69 SWU	1,392	33 per cent
Fabrication	1 kg	300	7 per cent

Spread across the 356 megawatt-hours that kilogram will generate, that is 11.89 dollars per megawatt-hour, or 1.19 cents per kilowatt-hour. At 2021 prices the same calculation gives 4.51 dollars. Fuel has become two and a half times more expensive in five years and remains a rounding error against the capital cost of the plant burning it, which is the durable economic fact about fission.

The last thing the unit is good for is something Dirac was probably not thinking about. The International Atomic Energy Agency defines a significant quantity as the approximate amount of material for which the possibility of building an explosive device cannot be excluded, and sets it at 25 kilograms of contained U-235 for highly enriched uranium and 75 kilograms for low-enriched [13]. Those are masses, in a ratio of three to one. But the barrier is separative work, and on that measure the two are nothing like three to one apart.

Twenty-five kilograms of contained U-235 at ninety per cent assay is 27.8 kilograms of metal, requiring 5,779 SWU from natural feed and about 5.4 tonnes of concentrate. Taken instead from 4.5 per cent reactor fuel it needs 1,281 SWU, and 655 kilograms of that fuel — well under the 75-kilogram-of-U-235 threshold at which the Agency's accounting begins to treat a stock as consequential. Reasoning in machines rather than tonnes is what the nonproliferation literature does with this arithmetic [15]: an AC100, rated above 340 SWU a year, would need seventeen machine-years to make one significant quantity from natural uranium. The demonstration cascade Centrus operates at Piketon, Ohio, contains sixteen centrifuges, and its obligation is at least 900 kilograms of 19.75 per cent HALEU a year [14]. Take

that obligation, assume 4.95 per cent feed, and the implied throughput is 5,300 SWU a year, or 331 SWU per machine — recovering the published machine rating from a production quota, two numbers never meant to be compared.

None of this is an argument against enriching uranium, which is how a fifth of the world's low-carbon electricity gets made. It is an argument for measuring the right thing. Assay is a percentage and reads like a distance still to be travelled; mass is intuitive and easy to inspect. Separative work is neither, and it is the only one of the three that tracks what a cascade actually has to do. Dirac's unpublished note picked the quantity that mattered, and eighty-five years later the industry cannot find a better one.

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